



FM Demodulators (Discriminators)

Complied By:
Kartik R. Patel
EXTC Department
KJSCE

FM Detectors Principle

- Coherent & Non-coherent
 - A coherent detector has **two inputs**—one for a reference signal, such as the synchronized oscillator signal, and one for the modulated signal that is to be demodulated.
 - A noncoherent detector has only **one input**, namely, the modulated signal port.
 - **Example:** The envelope detector is an example of a noncoherent detector.



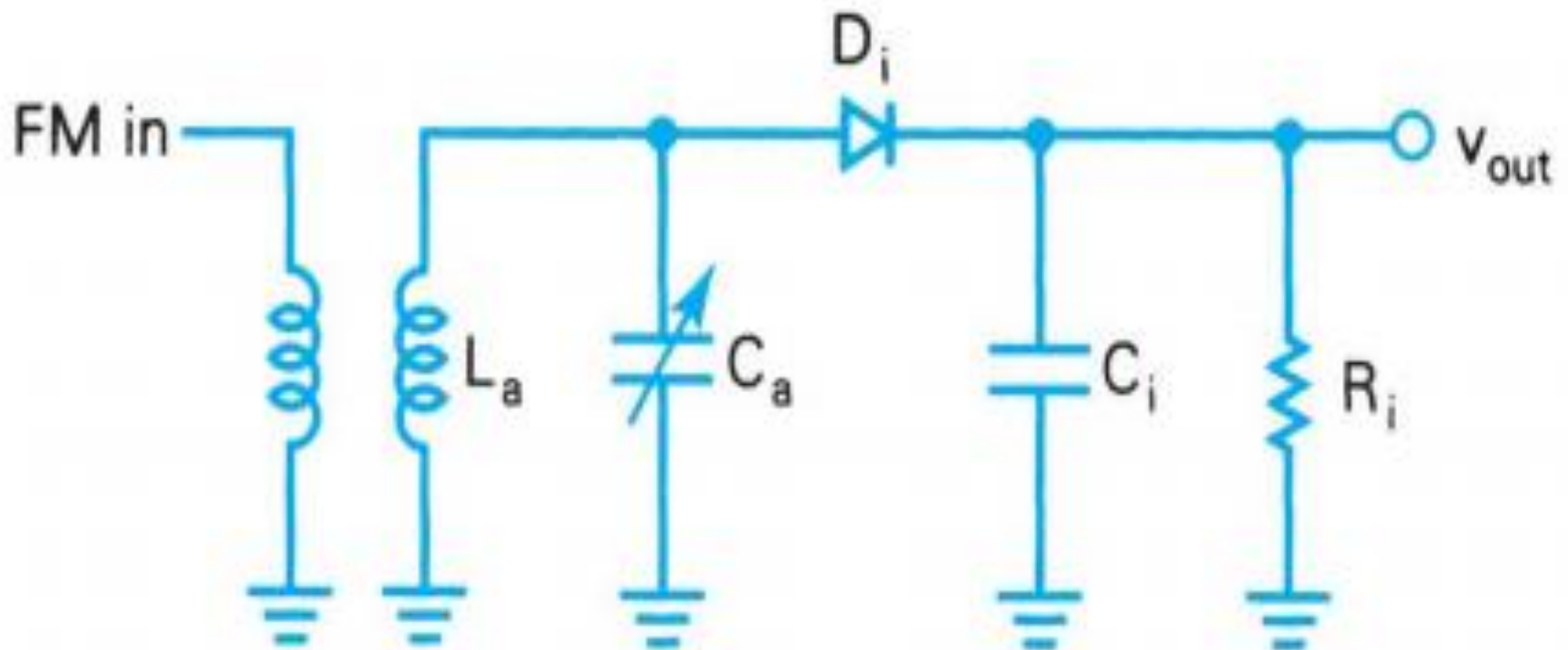
FM Detectors

- Demodulator Classification
 - Frequency Discrimination
 - Noncoherent demodulator
 - $FM \rightarrow AM \rightarrow ED \rightarrow m(t)$
 - Phase Shift Discrimination
 - Noncoherent demodulator
 - $FM \rightarrow PM \rightarrow m(t)$
 - Phase-Locked Loop (PLL) Detector
 - Coherent demodulator
 - Superior performance; complex and expensive

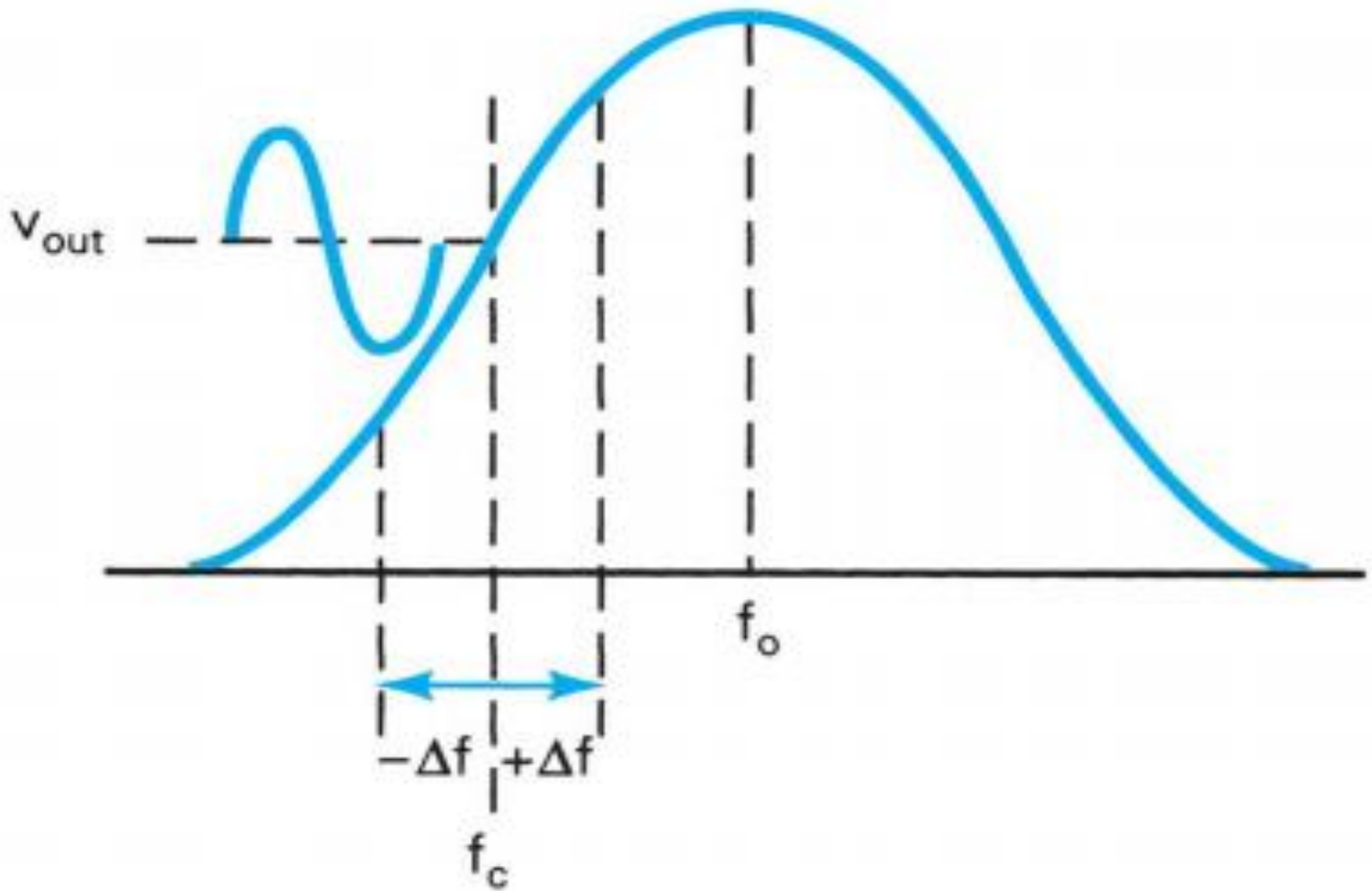
FM Detectors

- Function of frequency-to-amplitude changer is to change frequency deviations to amplitude variations.
- The conversion should be efficient and linear.
- Detection should be insensitive to amplitude changes.

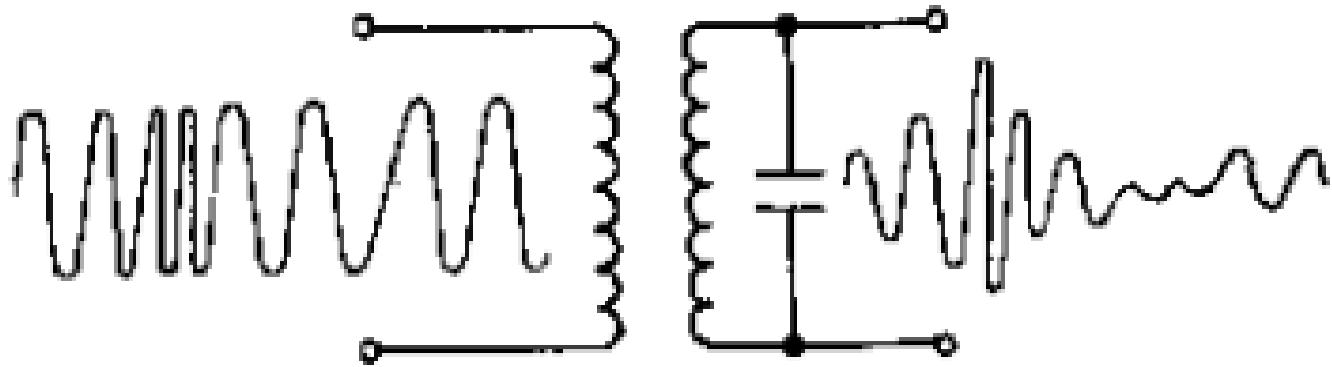
Slope Detection- Principle



Slope Detection- Principle



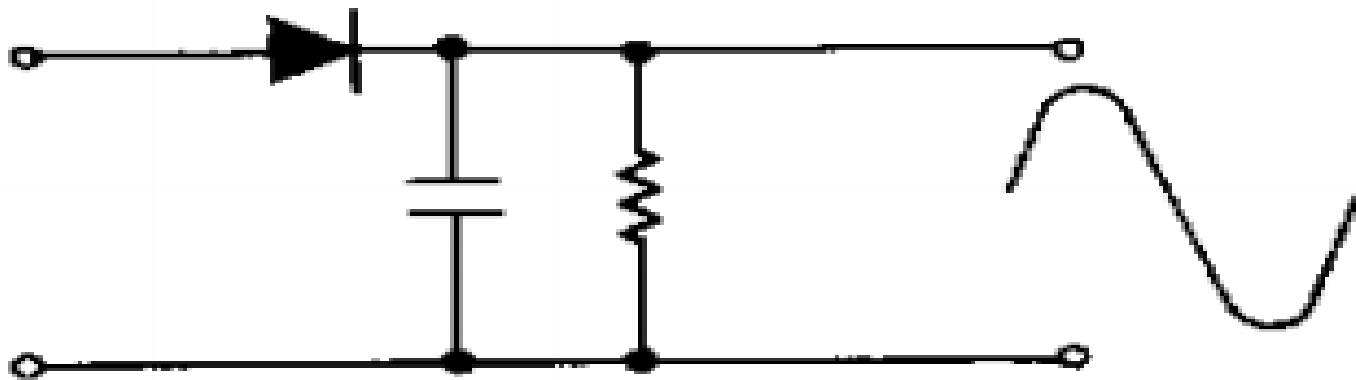
Slope Detection- Principle



(B) TANK CIRCUIT

$$K_d = \frac{V \text{ (Volts)}}{f \text{ (Hz)}}$$

$$V_{out}(t) = K_d \Delta f$$

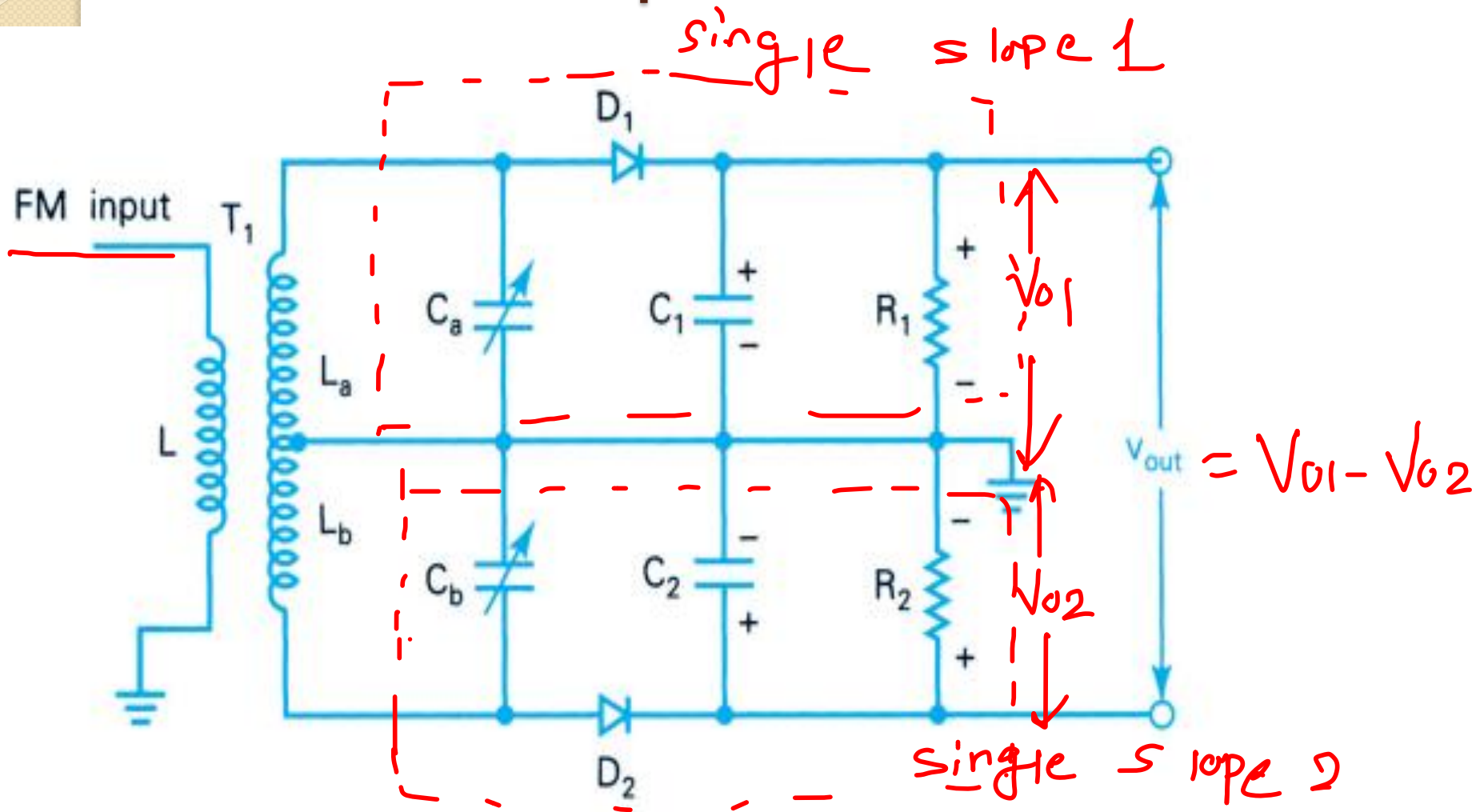


(C) DIODE DETECTOR

FM Detectors-Types

- Balance Slope Detector
- Foster-Seely Discriminator
- Ratio Detector
- FM Demodulation using PLL

I. Balanced Slope Detector



Case-I - $f_{in} = f_c$ (No deviation)

$$V_{01} = V_{02} \Rightarrow V_0 = 0$$

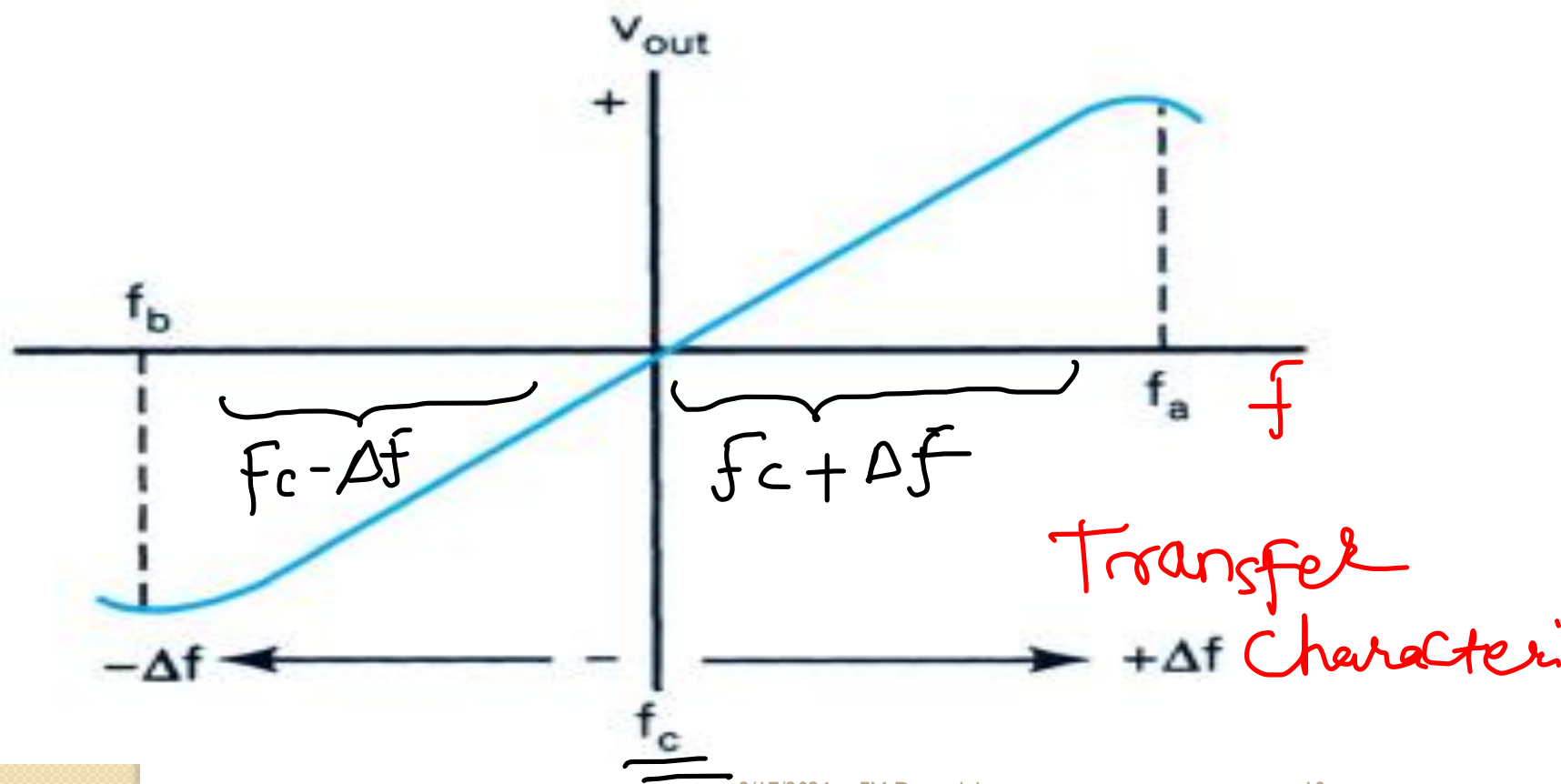
Case-II $\rightarrow f_{in} > f_c (f_c + \Delta f)$

$$V_{01} > V_{02} \Rightarrow V_0 = \text{Positive}$$

Case-III $\rightarrow f_{in} < f_c (f_c - \Delta f)$

$$V_{01} < V_{02} \Rightarrow V_0 = \text{Negative}$$

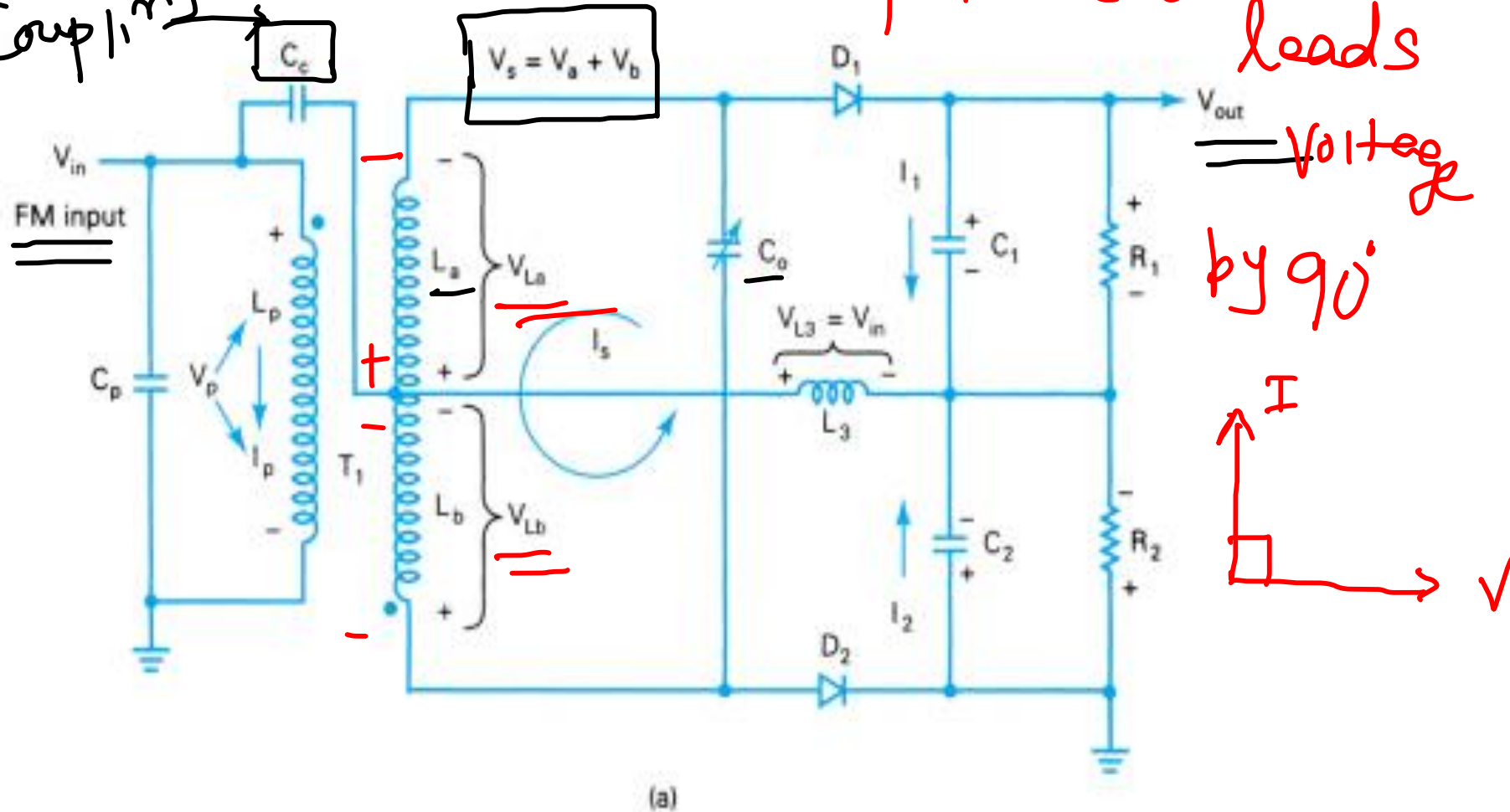
I. Balanced Slope Detector



2. Foster-Seely Discriminator

Coupling

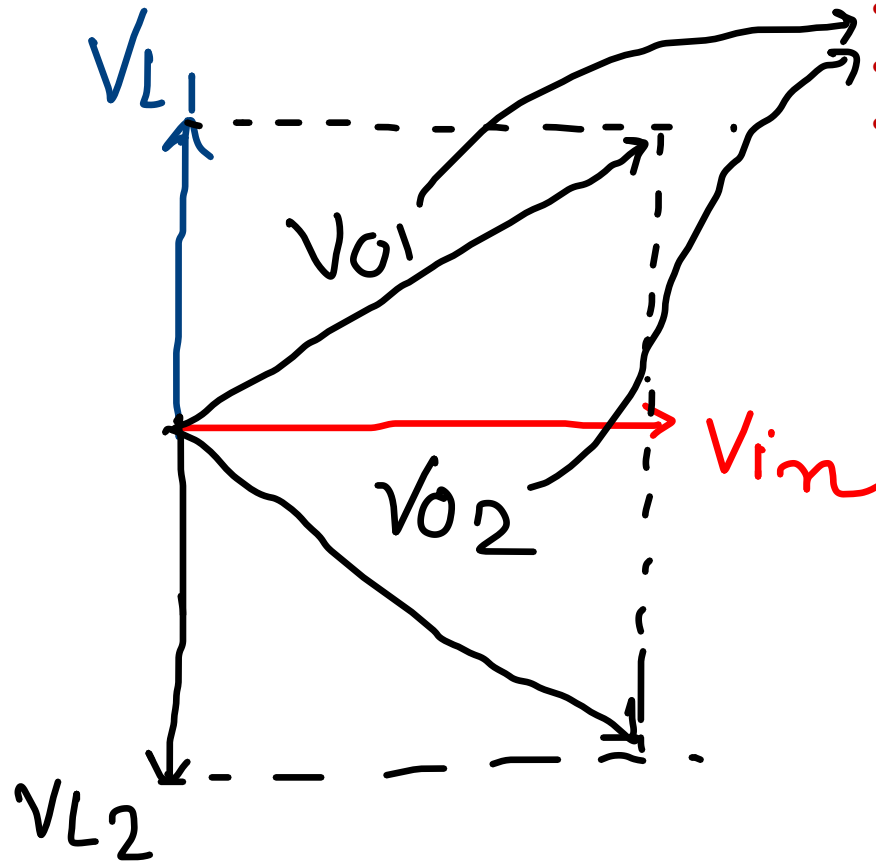
Capacitor current leads



2. Foster-Seely Discriminator

(no deviation) $V_{L1} = V_{L2}$

Case I: $f_{in} = f_c$



- Primary voltage: V_{in} ✓
- Secondary voltage: V_{L1} ✓
- $V_{o1} = V_{L1} + V_{in}$
- $V_{o2} = V_{L2} + V_{in}$
- $V_o(t) = V_{o1} - V_{o2}$

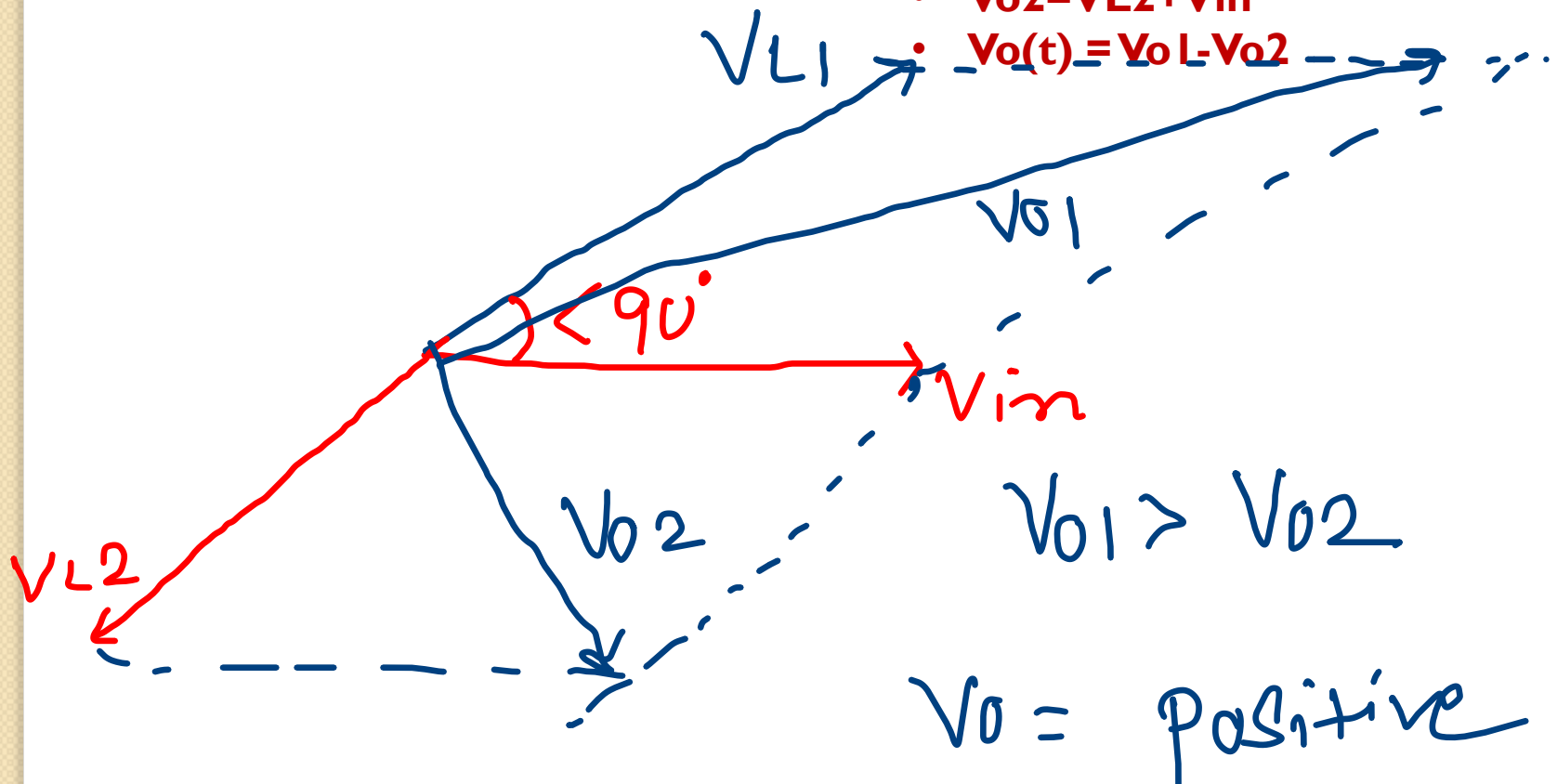
$$V_{L1} = V_{L2}$$

$$V_o = 0 \text{ V}$$

2. Foster-Seely Discriminator

Case 2: $f_{in} > f_c$ $(f_c + \Delta f)$

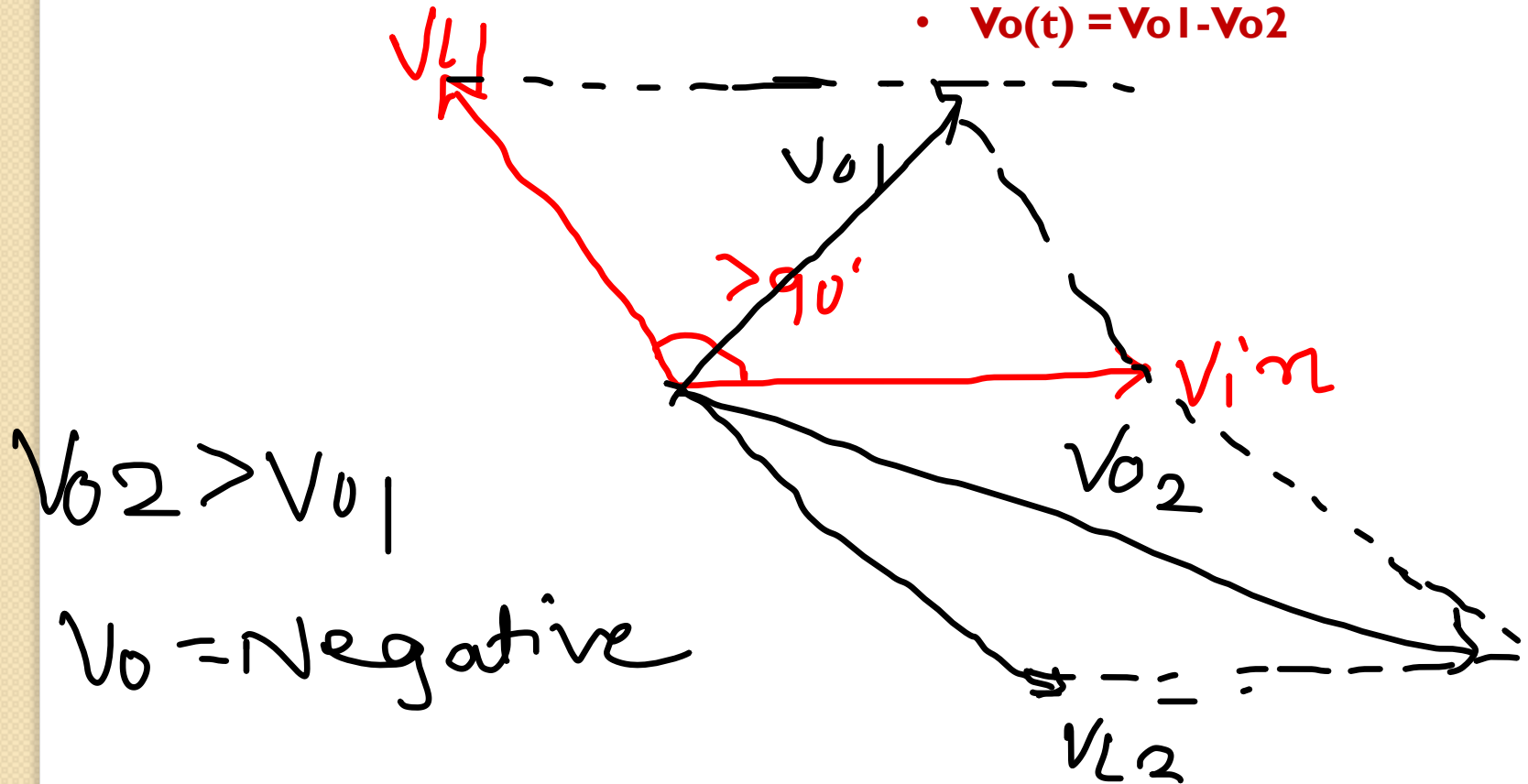
- Primary voltage: V_{in}
- Secondary voltage: V_{L1}
- $V_{o1} = V_{L1} + V_{in}$
- $V_{o2} = V_{L2} + V_{in}$
- $V_o(t) = V_{o1} - V_{o2}$



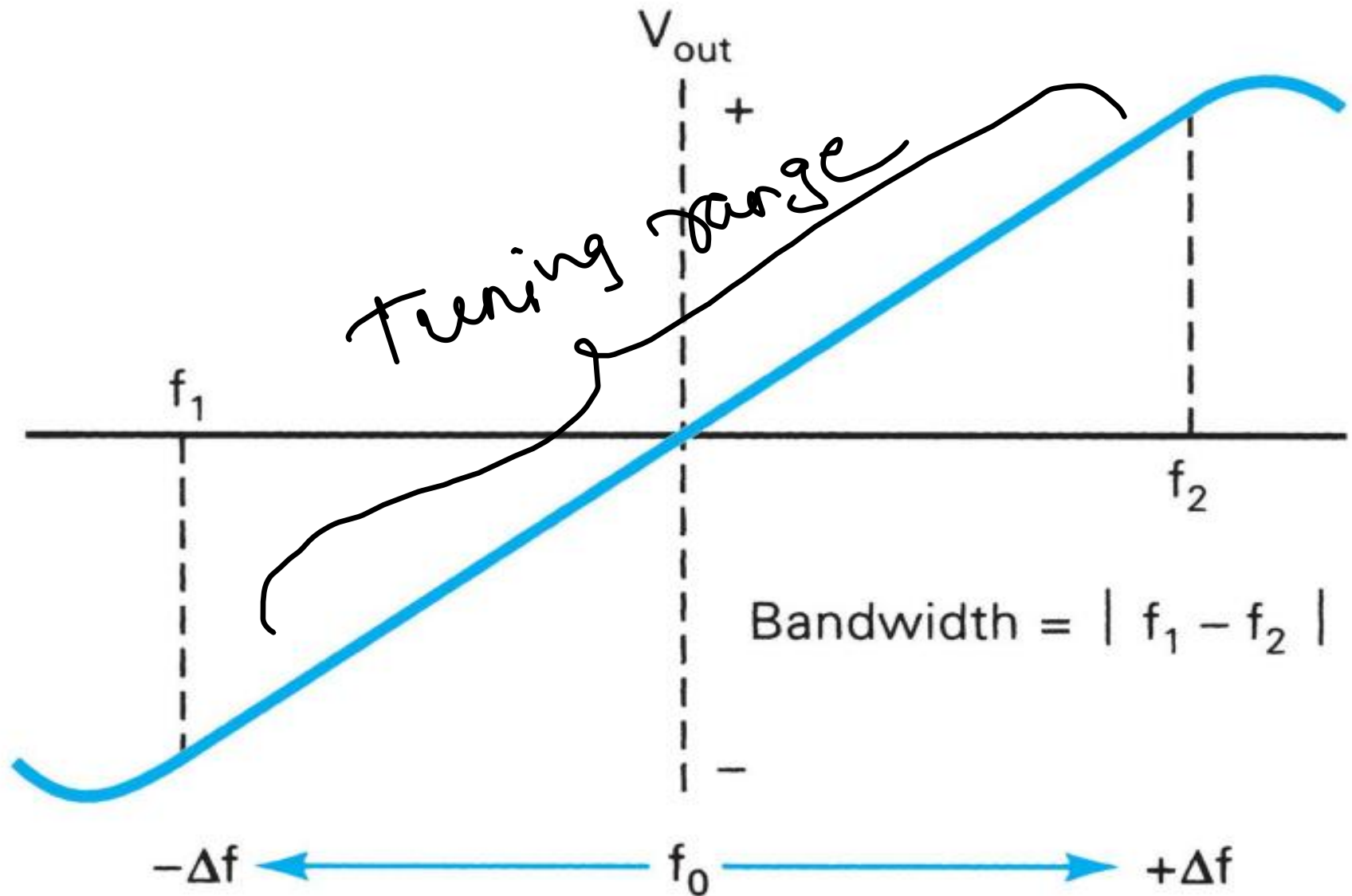
2. Foster-Seely Discriminator

Case 3: $f_{in} < f_c$ $(f_c - \Delta f)$

- Primary voltage: V_{in}
- Secondary voltage: V_{L1}
- $V_{o1} = V_{L1} + V_{in}$
- $V_{o2} = V_{L2} + V_{in}$
- $V_o(t) = V_{o1} - V_{o2}$



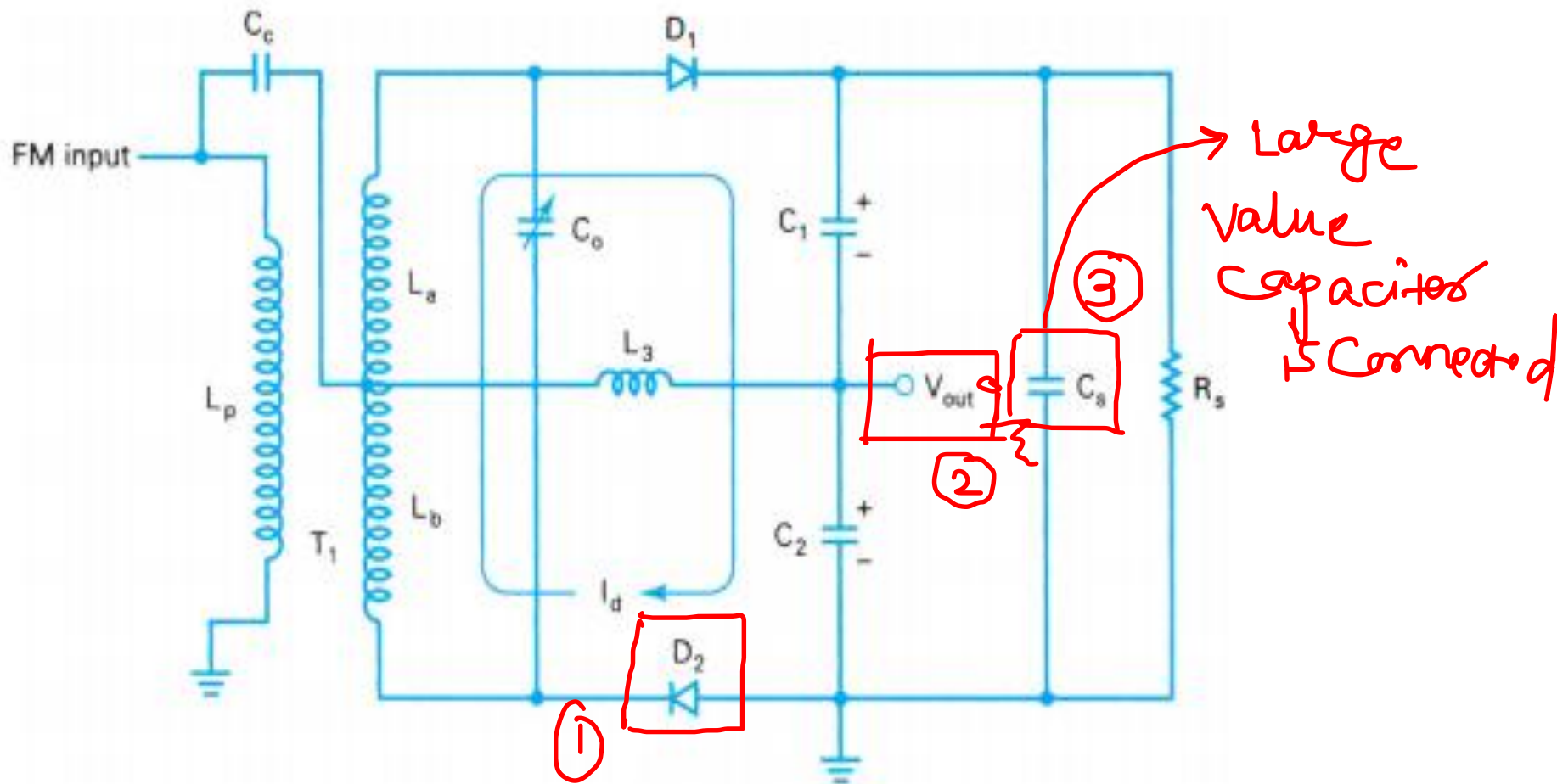
2. Foster-Seely Discriminator



Disadvantage of Foster-seely discriminator

- Weak AM signal (too small in amplitude to reach limiting level), *Noise*
- Passes through the limiter stages, and can appear in output.
- Small AM signals are detected by the diode detector.
- Problem is solved in ratio detector.

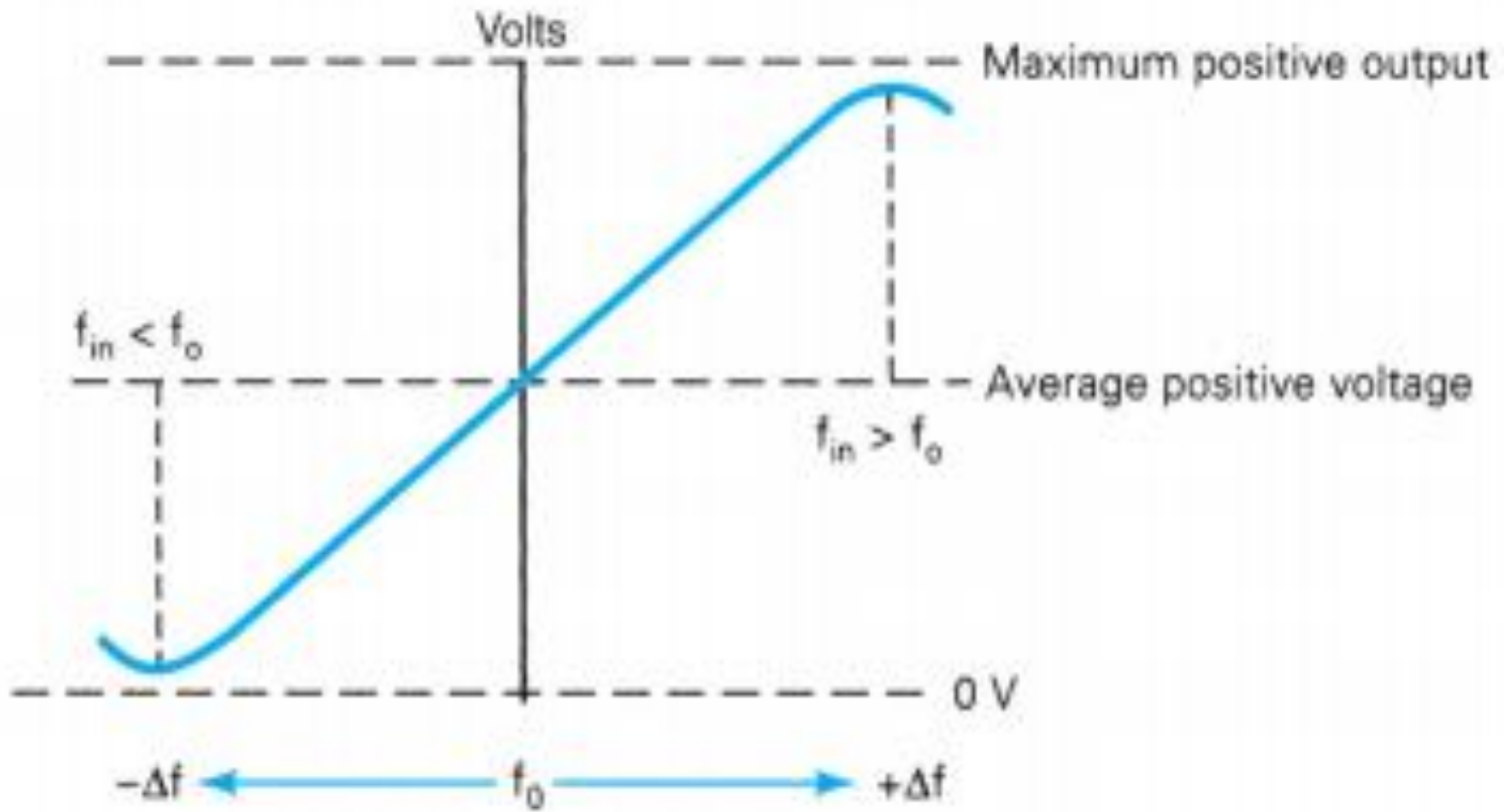
3. Ratio Detector



Changes in Ratio Detector

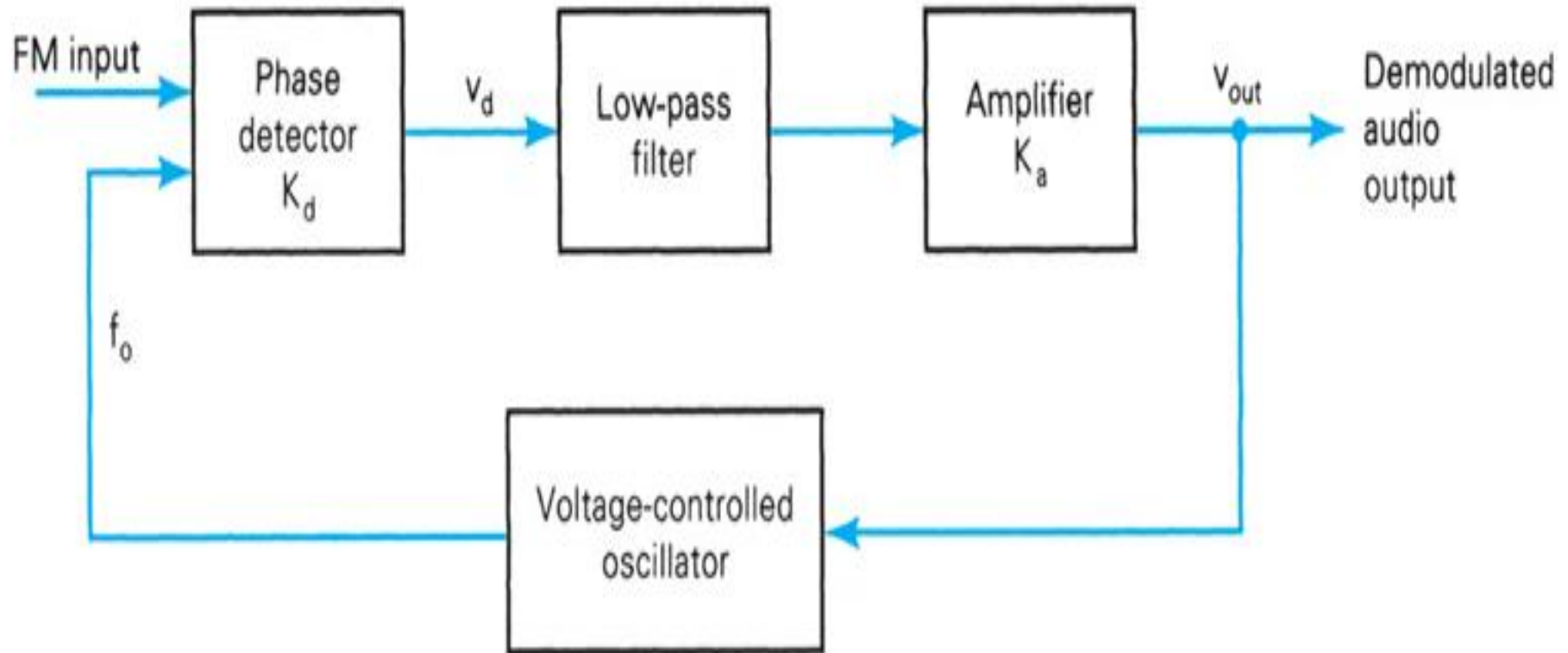
- Position of diode D2 is reversed.
- Output is taken from center tapping of transformer.
- A large value capacitor C5 is added for amplitude limiting.

3. Ratio Detector



(b)

4. PLL FM Demodulator



(a)